POLITECNICO DI MILANO

High Performance Computing Engineering

Quantum Physics – September 12, 2023

Please justify each answer and use a clear handwriting. Define each symbol you employ. For each exercise, give the symbolic solution and substitute numerical values (if required) only at the end. Write your first name, family name, and personal code on each sheet in a readable way.

1. A particle of mass m can move only along the x axis and is subject to a harmonic force characterized by an elastic constant equal to k.

a) Express the Hamiltonian operator of the particle as a function of the ladder operators and find the position and momentum expectation values of the particle for any eigenstate of the Schrödinger equation.

b) Explicitly evaluate the standard deviations associated with the measurements of the particle position and momentum for any eigenstate.

c) Assume that, at time $t = 0$, the particle is described by the ket $|u\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$, where $|0\rangle$ and $|1\rangle$ are the oscillator ground state and the first excited state, respectively. Evaluate the expectation value of the particle position as a function of time.

d) At a certain time, the particle position is measured and the particle is found at position x_0. What is the probability that the system will be found in the ground state *after* the measurement of the particle position?

[*Suggestion*: in position representation, $|0\rangle = \psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}x^2}$]

2. The spin state of an electron is described by the ket $|\chi\rangle = \frac{1}{\sqrt{5}}(2|\uparrow_z\rangle + i|\downarrow_z\rangle)$, with $\hat{s}_z|\uparrow_z\rangle = \frac{\hbar}{2}|\uparrow_z\rangle$ and $\hat{s}_z|\downarrow_z\rangle = -\frac{\hbar}{2}|\downarrow_z\rangle$, where $\hat{s}_z$ is the projection of the spin operator along the z axis.

a) Find the expectation value of the electron spin (remember that spin is a vector!).

b) List all the possible values of the outcomes of the measurements of the spin projection either along x, y, or z, giving, for each value, its probability.

c) Evaluate the expectation value of the commutator $[\hat{s}_x, \hat{s}_y]$ for the ket $|\chi\rangle$ and discuss how it is related to the lower limit of the product of the standard deviations $\sigma_{s_x}\sigma_{s_y}$ associated with a combined measurement of $\hat{s}_x$ and $\hat{s}_y$.

d) When the electron is immersed in a uniform magnetic field $\mathbf{B}$ directed along z, its spin dynamics is governed by the Hamiltonian $\hat{H} = -\frac{q}{m}B\hat{s}_z$, where $\frac{q}{m}$ is the (negative) electron charge-to-mass ratio. Determine the expectation value of the x component of the electron spin as a function of time.

Solution to the 1st exercise

a) $\hat{H} = \hbar\omega\left(\hat{a}\hat{a}^\dagger - \frac{1}{2}\right) = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right) = \frac{1}{2}\hbar\omega(\hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger)$, with $\omega = \sqrt{\frac{k}{m}}$.

The eigenstates are $|n\rangle$, with $\hat{H}|n\rangle = \hbar\omega\left(n + \frac{1}{2}\right)|n\rangle$.

To find the particle position and momentum expectation values it is convenient to express the corresponding operators as a function of the ladder operators:

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}^\dagger + \hat{a}), \qquad \hat{p} = i\sqrt{\frac{m\omega\hbar}{2}}(\hat{a}^\dagger - \hat{a}).$$

From these relations one finds:

$$\langle n|\hat{x}|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\langle n|\hat{a}^\dagger + \hat{a}|n\rangle = 0, \qquad \langle n|\hat{p}|n\rangle = i\sqrt{\frac{m\omega\hbar}{2}}\langle n|\hat{a}^\dagger - \hat{a}|n\rangle = 0.$$

b) $\sigma_x^2 = \langle n|\hat{x}^2|n\rangle - \langle n|\hat{x}|n\rangle^2 = \langle n|\hat{x}^2|n\rangle = \frac{\hbar}{2m\omega}\left\langle n\left|\left(\hat{a}^\dagger + \hat{a}\right)^2\right|n\right\rangle = \frac{\hbar}{2m\omega}\langle n|\hat{a}^\dagger\hat{a}^\dagger + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + \hat{a}\hat{a}|n\rangle$
$= \frac{\hbar}{2m\omega}\langle n|\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}|n\rangle = \frac{\hbar}{2m\omega}(2n+1).$

$\sigma_p^2 = \langle n|\hat{p}^2|n\rangle - \langle n|\hat{p}|n\rangle^2 = \langle n|\hat{p}^2|n\rangle = -\frac{m\omega\hbar}{2}\left\langle n\left|\left(\hat{a}^\dagger - \hat{a}\right)^2\right|n\right\rangle =$
$= -\frac{m\omega\hbar}{2}\langle n|\hat{a}^\dagger\hat{a}^\dagger - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} + \hat{a}\hat{a}|n\rangle = \frac{m\omega\hbar}{2}\langle n|\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}|n\rangle = \frac{m\omega\hbar}{2}(2n+1).$

c) $|u(t)\rangle = \frac{1}{\sqrt{2}}\left(e^{-i\frac{1}{2}\omega t}|0\rangle + ie^{-i\frac{3}{2}\omega t}|1\rangle\right)$

$$\Rightarrow \quad \langle u(t)|\hat{x}|u(t)\rangle = \sqrt{\frac{\hbar}{2m\omega}}\langle u(t)|\hat{a}^\dagger + \hat{a}|u(t)\rangle = \sqrt{\frac{\hbar}{8m\omega}}\left(-ie^{i\omega t}\langle 1|\hat{a}^\dagger|0\rangle + ie^{-i\omega t}\langle 0|\hat{a}|1\rangle\right)$$
$$= \sqrt{\frac{\hbar}{8m\omega}}\left(-ie^{i\omega t} + ie^{-i\omega t}\right) = \sqrt{\frac{\hbar}{2m\omega}}\sin(\omega t).$$

d) In this case, it is convenient to use the position representation, in which $|0\rangle = \psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}x^2}$.
Because of the position measurement, the wavefunction collapses on the pure x_0 state $|x_0\rangle = \delta(x - x_0)$, with $\delta(x)$ being the Dirac delta function.
Therefore, the probability $\mathcal{P}$ that the particle is found in the ground state after the position measurement is given by:

$$\mathcal{P} = |\langle x_0|0\rangle|^2 = \left|\left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}}\int_{-\infty}^{\infty} e^{-\frac{m\omega}{2\hbar}x^2}\delta(x - x_0)dx\right|^2 = \sqrt{\frac{m\omega}{\pi\hbar}}e^{-\frac{m\omega}{\hbar}x_0^2} = |\psi_0(x_0)|^2.$$

Solution to the 2nd exercise

a) $\langle \mathbf{s} \rangle = \langle \chi | \hat{\mathbf{s}} | \chi \rangle = \langle \chi | \hat{s}_x | \chi \rangle \mathbf{u}_x + \langle \chi | \hat{s}_y | \chi \rangle \mathbf{u}_y + \langle \chi | \hat{s}_z | \chi \rangle \mathbf{u}_z = \frac{\hbar}{2} \left(\langle \chi | \hat{\sigma}_x | \chi \rangle \mathbf{u}_x + \langle \chi | \hat{\sigma}_y | \chi \rangle \mathbf{u}_y + \langle \chi | \boldsymbol{\sigma}_z | \chi \rangle \mathbf{u}_z \right)$,

with $\hat{\sigma}_x$, $\hat{\sigma}_y$, and $\hat{\sigma}_z$ being the Pauli matrices and $\mathbf{u}_x$, $\mathbf{u}_y$, and $\mathbf{u}_z$ the unit vectors of the Cartesian axes. By substituting, the corresponding expressions, we obtain:

$$\langle \mathbf{s} \rangle = \frac{\hbar}{2} \left(\frac{4}{5} \mathbf{u}_y + \frac{3}{5} \mathbf{u}_z \right).$$

Note that $|\langle \mathbf{s} \rangle| = \frac{\hbar}{2}$.

b) From the expression of $|\chi\rangle$ we can immediately deduce that the outcomes of a measurement of $\hat{s}_z$ are:

$s_z = \frac{\hbar}{2}$ with probability $\mathcal{P}\left(s_z = \frac{\hbar}{2}\right) = \frac{4}{5}$, $\qquad s_z = -\frac{\hbar}{2}$ with $\mathcal{P}\left(s_z = -\frac{\hbar}{2}\right) = \frac{1}{5}$

To evaluate the outcomes of the measurements of the spin projection either along x or y we need to know the eigenstates and eigenvalues of $\hat{s}_x$ and $\hat{s}_y$:

$\hat{s}_x |\uparrow_x\rangle = \frac{\hbar}{2} |\uparrow_x\rangle$, with $|\uparrow_x\rangle = \frac{1}{\sqrt{2}} (|\uparrow_z\rangle + |\downarrow_z\rangle)$, and $\quad \hat{s}_x |\uparrow_x\rangle = -\frac{\hbar}{2} |\uparrow_x\rangle$, with $|\uparrow_x\rangle = \frac{1}{\sqrt{2}} (|\uparrow_z\rangle - |\downarrow_z\rangle)$.

$\hat{s}_y |\uparrow_y\rangle = \frac{\hbar}{2} |\uparrow_y\rangle$, with $|\uparrow_y\rangle = \frac{1}{\sqrt{2}} (|\uparrow_z\rangle + i|\downarrow_z\rangle)$, and $\quad \hat{s}_y |\uparrow_y\rangle = -\frac{\hbar}{2} |\uparrow_y\rangle$, with $|\uparrow_y\rangle = \frac{1}{\sqrt{2}} (|\uparrow_z\rangle - i|\downarrow_z\rangle)$.

From these expressions we obtain:

$s_x = \frac{\hbar}{2}$ with $\mathcal{P}\left(s_z = \frac{\hbar}{2}\right) = |\langle \uparrow_x | \chi \rangle|^2 = \frac{1}{2}$, $\qquad s_x = -\frac{\hbar}{2}$ with $\mathcal{P}\left(s_x = -\frac{\hbar}{2}\right) = |\langle \downarrow_x | \chi \rangle|^2 = \frac{1}{2}$;

$s_y = \frac{\hbar}{2}$ with $\mathcal{P}\left(s_y = \frac{\hbar}{2}\right) = \left|\langle \uparrow_y | \chi \rangle\right|^2 = \frac{9}{10}$, $\qquad s_y = -\frac{\hbar}{2}$ with $\mathcal{P}\left(s_y = -\frac{\hbar}{2}\right) = \left|\langle \downarrow_y | \chi \rangle\right|^2 = \frac{1}{10}$.

c) $[\hat{s}_x, \hat{s}_y] = i\hbar \hat{s}_z \quad \Rightarrow \quad \langle [\hat{s}_x, \hat{s}_y] \rangle = i\hbar \langle s_z \rangle = \frac{3}{10} \hbar^2 \quad \Rightarrow \quad \sigma_{s_x} \sigma_{s_y} \geq \frac{1}{2} \left| \langle [\hat{s}_x, \hat{s}_y] \rangle \right| = \frac{3}{20} \hbar^2$.

From the values obtained in the previous point, one indeed obtains $\sigma_{s_x} \sigma_{s_y} = \frac{3}{20} \hbar^2$.

d) In the representation of the s_z eigenvalues, the time-evolution operator assumes the following matrix form:

$$\hat{T} = \begin{bmatrix} e^{i\frac{q_e}{2m_e}Bt} & 0 \\ 0 & e^{-i\frac{q_e}{2m_e}Bt} \end{bmatrix}.$$

Therefore,

$$\langle s_x(t) \rangle = \langle \chi | \hat{T}^{-1} \hat{s}_x \hat{T} | \chi \rangle = \frac{\hbar}{10} [2 \quad -i] \cdot \begin{bmatrix} 0 & e^{-i\frac{q}{m}Bt} \\ e^{i\frac{q}{m}Bt} & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ i \end{bmatrix} = \frac{2}{5} \hbar \sin\left(\frac{Bq}{m} t\right).$$